Smaller Diophantine forms of ZFC consistency

A twelve-variable polynomial of degree 1081 in 116 operations, with a fivefold smaller coefficient-base exponent

OpenAI-assisted research synthesis prepared for
Vladimir Reshetnikov's ProveIt repository

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Comparison snapshot: `ae3007ca4467`

Abstract

We compare the eight current packages in ProveIt's third collection of Diophantine formulations of Con(ZFC) and combine two of their root elimination methods with a smaller coefficient encoding. The new ordinary integer polynomial has twelve strictly positive unknowns, exact total degree 1081, and a shared arithmetic circuit with 116 binary operations. It improves the inspected twelve-variable degree record 1145 at 123 operations and matches the 116-operation construction of degree 1353. Its three-equation presentation has maximum degree 1072 and 111 operations under the collection's free-comparison convention.

Separately, mixed-radix clause decoding and external zero tests reduce the 943-tile compiler base from $2^{5^{10}}$ to 2^{5^9}. All four coefficients are materialized: their combined raw size falls from 5 312 269 to 977 220 bytes. This tuple also instantiates the existing 67-operation polynomial and 59-operation system, included with unchanged arithmetic.

The proof composes explicit new algebraic and number-theoretic arguments with the identified native compiler and K215 consistency calibration. It is ordinary mathematics formalizable in ZFC, supported by reproducible finite audits, and is not a new end-to-end Lean or Rocq certificate. No consistency proof, global minimum, or improvement on the existing nine-natural-variable count is asserted.

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1 The result and the meaning of simplicity

The central deliverable is a completely specified polynomial $\mathcal{P}_{12} \in \mathbb{Z}[J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r]$. Its four coefficient ports are bound to four particular integers, not existential parameters. The full definition appears in Sections 2 and 6; executable binary circuits and the actual coefficient bytes accompany this article.

Theorem 1.1 (Fixed-instance equivalence)**.** *For the coefficient tuple defined in Section 6, ZFC proves*

$$\boxed{(\nexists \mathbf{x} \in \mathbb{Z}_{>0}^{12} : \mathcal{P}_{12}(\mathbf{x}) = 0) \iff \mathrm{Con(ZFC)}.}$$

The three-equation system and sum-of-squares equation in Section 2 have the same existence predicate. The polynomial has exact degree 1081 *and a* 116*-operation circuit; the system has maximum residual degree* 1072 *and a* 111*-operation circuit.*

Here Con(ZFC) means absence of a finite syntactic derivation of falsity in the effective first-order ZFC presentation used by the K215 endpoint. The theorem proves an equivalence. It assumes neither side of that equivalence. The twelve-coordinate domain is strictly positive integers. For nonnegative unknowns one may write $\mathcal{P}_{12}(n_1 + 1, \ldots, n_{12} + 1) = 0$; its degree remains 1081. The displayed equation is not asserted equivalent over unrestricted $\mathbb{Z}^{12}$.

1.1 Comparison at an immutable snapshot

The comparison is frozen at `ae3007ca4467011e0e051c84acb4a1dd13971b0e`. The current collection has packages 16–23; its earlier waves survive in history and in cited source copies [1]. The main measured tradeoffs are:

Construction	Unknowns	Equations	Degree	Operations
18: Jones specialization	$9\mathbb{N}$	1	$161844\,5^{33} + 484$	20,531
20: previous lowest-degree twelve	$12\mathbb{Z}_{>0}$	1	1145	123
21: previous least-operation twelve	$12\mathbb{Z}_{>0}$	1	1353	116
22: monic twelve	$12\mathbb{Z}_{>0}$	1	1529	117
New hybrid	$\mathbf{12\mathbb{Z}_{>0}}$	**1**	**1081**	**116**
New hybrid system	$\mathbf{12\mathbb{Z}_{>0}}$	**3**	**1072**	**111**
New hybrid sum of squares	$12\mathbb{Z}_{>0}$	1	2144	119
Inherited least-operation polynomial	$15\mathbb{Z}_{>0}$	1	155	67
22: inherited least-operation system	$16\mathbb{Z}_{>0}$	7	24	59
Inherited quartic lift	$32\mathbb{Z}_{>0}$	1	4	—

The degree and operation improvement is simultaneous within the reviewed twelve-variable family: $1081 < 1145 < 1353$ and $116 < 123$. Relative to package 20 the degree falls by 64, approximately 5.59%; relative to package 21 it falls by 272, approximately 20.10%. These statements do not compare unlike domains or imply global optimality.

The 111-operation system requires an accounting qualification. Our emitter compares the two final norm terms directly instead of subtracting them and comparing with zero. The same rewrite reduces package 21's published 112-operation system to 111 operations. Thus the new system has lower degree at a tied normalized cost. The apparent one-operation gain is not exclusive to this construction.

An operation means one binary addition, subtraction, or multiplication, including a square and multiplication by a fixed numeral. Register reuse, supplied fixed coefficients, and equality comparisons are free. These are costs of the supplied circuits, not lower bounds on all possible circuits. We report maximum residual degree for systems and total degree for a single polynomial.

1.2 Which presentation to use

For a compact moderate-degree formula with few coordinates, the new twelve-variable product is the strongest result here. For arithmetic operation count alone, the inherited 67-operation product remains better; for a small system, retain the 59-operation presentation. Both are included with the new, smaller coefficient tuple. If coordinate count alone is decisive, package 18's nine-natural-variable specialization remains smaller, at much larger degree and circuit size [10].

The collection also contains 26-integer and 27-integer constructions, including a 27-integer product of degree 172 [1]. Those are different domain tradeoffs. None of the new positive-domain arguments licenses silently interpreting our twelve coordinates as signed.

2 A complete polynomial definition

Supply exactly the twelve positive integers

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r.$$

Write the four fixed ports as β, K_0, M_C, M_F. The last port is the *shifted* mask $M_F = M_F^{\mathrm{native}} + B - 1$. All other symbols below are polynomial abbreviations, not additional unknowns.

2.1 The common arithmetic interface

Define

$$\begin{aligned}
&q = \beta J + 1, \quad C_w = q - F - \alpha, && X = wq, && Y = sq^3,\\
&E = XY, && k = \eta + \zeta, && c = kY + \eta, && a = E + Y,\\
&H = 4a + 3, && \Delta = a^2 + H = (a+2)^2 - 1,\\
&\ell = EkY, && A = 1 + \ell(\ell + k), && G = 1 + \Delta c^2, && b_0 = X + ac.
\end{aligned}$$

The packed rank and outer factors are

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \tag{1}$$
$$N_k = k - hE - R, \tag{2}$$
$$N_t = (K_0 + w)C_w + (q - F) - z(q-1). \tag{3}$$

For the strong and auxiliary Pell constraints, put

$$\begin{aligned}
&I = ic^2, && D = 1 + \Delta I^2, && Q = (\Delta I)^2, && d = Q - 1,\\
&v = cT, && b = c + RD, && P = QDv^2, && U = Qb^2,\\
&e = P + U - 1, && B_{\mathrm{aux}} = 2Qvb.
\end{aligned}$$

Our new choice of weights and carrier is

$$\boxed{M = (kI + T)^2, \qquad L = 1 + HM, \qquad S = b_0 + Hr.} \tag{4}$$

2.2 A conceptual norm definition

For a temporary indeterminate x, define

$$\begin{aligned}
C(x) &= d(x^2 + M^2 A) - e,\\
\mathcal{F}(x) &= \left(C(x)^2 + DB_{\mathrm{aux}}^2 - 4d^2x^2M^2A\right)^2 - 4DB_{\mathrm{aux}}^2 C(x)^2.
\end{aligned}$$

Let $\mathcal{F}_H(t) = H^8\mathcal{F}(t/H)$. The denominator clears because $\deg_x \mathcal{F} = 8$ and all its coefficients are integers. Define the norm

$$\mathcal{N} = \mathcal{F}_H(S - L\sqrt{G})\,\mathcal{F}_H(S + L\sqrt{G}). \tag{5}$$

The two conjugate factors make this an integer polynomial. No radical, division, variable exponent, or quantified auxiliary root occurs in the actual evaluator supplied next.

2.3 A division-free scalar evaluator

The following formulas eliminate all formal radicals explicitly:

$$\begin{aligned}
n &= (HM)^2A, & V_0 &= S^2 + L^2G, & V_1 &= 2SL, \\
C_0 &= d(V_0 + n) - H^2e, & C_1 &= dV_1, & & \\
K_f &= 4d^2n, & K_{c0} &= K_fV_0, & K_{c1} &= K_fV_1, \quad Z = 4H^4PU, \\
C_{s0} &= C_0^2 + GC_1^2, & C_{s1} &= 2C_0C_1, & & \\
W_0 &= C_{s0} - K_{c0} - Z, & W_1 &= C_{s1} - K_{c1}, & & \\
N_0 &= W_0^2 + GW_1^2 - 4ZK_{c0}, & N_1 &= 2W_0W_1 - 4ZK_{c1}, & & \\
\mathcal{N} &= N_0^2 - GN_1^2. & & & &
\end{aligned}$$

The fixed coefficient K_0 and the intermediate K_{c0} are different symbols. This distinction is also enforced in the JSON circuits. The identity used to save operations is

$$(C^2 + K - Z)^2 - 4C^2K = (C^2 - K - Z)^2 - 4KZ.$$

It replaces a general multiplication of two quadratic coefficient pairs by multiplication of a pair by a scalar. The identity is inherited from package 21 [3]; the distinct inner and outer weights are new.

The recommended single equation is

$$\boxed{\mathcal{P}_{12} = N_kN_t(1 + \mathcal{N}) - 1 = 0.} \tag{6}$$

The equivalent three-equation presentation is

$$\boxed{N_k = 1, \qquad N_t = 1, \qquad N_0^2 = GN_1^2.} \tag{7}$$

The conservative sum-of-squares alternative is

$$(N_k - 1)^2 + (N_t - 1)^2 + \mathcal{N}^2 = 0. \tag{8}$$

Its degree is 2144 and its circuit uses 119 operations.

3 The logical interface and proof dependencies

Let $K215$ denote the literal machine included as `coefficients/k215.tm`. Its working states are $0, \ldots, 214$; state 215 is halt. The initial tape is blank, with the head in state zero. The table SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The pinned endpoint states

$$\mathrm{BlankHalts}(K215) \iff \mathrm{BProv}(\mathrm{ZFCax}_s, [], \bot)[11].$$

We use the ordinary mathematical calibration mirrored by this endpoint, not a meta-inference from Lean's soundness to a ZFC proof object. No endpoint build was rerun for this delivery.

The inherited normalized blank compiler has positive coordinates

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma$$

and the relations

$$\begin{aligned} &\tau^2 = A, && u = X + ac + H\gamma, \qquad u^2 = G, \\ &f^2 = 1 + \Delta i^2 c^4 = D, \qquad && Q(c(Tf-1) - Rf^2)^2 - (Q-1)y^2 = 1, \\ &N_k = 1, && N_t = 1. \end{aligned}$$

Its signed-quotient extension permits the simultaneous replacement $(f, T) \mapsto (-f, -T)$ during reconstruction; the detailed source arguments are included in `upstream/`. This extension, the Pell rank and half-binomial lemmas, the native compiler, and K215 calibration are inherited dependencies [5, 2, 4, 6].

The new work has two independent parts: prove that the twelve-variable norm restores exactly this arithmetic certificate, and prove that the new coefficient tuple meets the same native compiler interface. Together they establish

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ \mathcal{P}_{12} = 0 \iff \mathrm{BlankHalts}(K215) \iff \neg\,\mathrm{Con}(\mathrm{ZFC}).$$

All statements concern finite strings, natural-number arithmetic, algebraic numbers, or elementary Pell theory, and the cited constructions are formalizable in ZFC. The present package does not contain a newly checked end-to-end formalization of this chain.

4 The new root-recovery proof

4.1 Unit signs and the noncircular bootstrap

In the scalar evaluator, V_1 and C_1 are even, C_{s1} is divisible by 4, K_{c1} by 8, W_1 by 4, and N_1 by 8. Consequently

$$\mathcal{N} \equiv N_0^2 \pmod{64}.$$

At an integer zero of (6), the three factors $N_k, N_t, 1+\mathcal{N}$ are units. The value $1+\mathcal{N} = -1$ would make a square congruent to -2 modulo 4, which is impossible. Therefore

$$\mathcal{N} = 0, \qquad N_k = N_t = \varepsilon \in \{-1, 1\}. \tag{9}$$

The inherited weak transport estimate applies before any root recovery [5]. Since $K_0 + w \geq 2$, the transport factor being a unit implies $C_w \geq 0$. In particular $q - F = C_w + \alpha > 0$. For the coherent compiler tuple its mask bounds then give

$$\begin{aligned} &q \geq 16, \quad X \geq q, \quad Y \geq q^3 \geq 4096, \quad 0 < R < q^4 - 2, \\ &E \geq q^4 > R + 2, \quad k = R + \varepsilon + hE > E, \quad k > R, \quad kY < c < k(Y+1) < 2kY. \end{aligned} \tag{10}$$

The exceptional boundary $C_w = 0$ is retained at this stage. It is not discarded by assuming the final index sign prematurely. Set $A_0 = a + 2$. Since $E \geq 16Y$ and $Y \geq 4096$, we also have $A_0 < 2E$.

4.2 The inner weight needs no extra doubling

Put $P_0 = Dv^2 + b^2$ and $M_0 = kI^2 + T^2$. The estimates (10) give

$$A > \frac{A_0^2 c^2}{16}, \qquad A > 64A_0^4, \qquad D < A_0^2 I^2, \qquad b < 2kA_0^2 I^2. \tag{11}$$

For the first two, $A > \ell^2$, $A_0 c < 4EkY = 4\ell$, and $\ell > E^2 Y$ suffice. The second follows from $E^4 Y^2 > 64(2E)^4$. The remaining bounds use $I > 1$, $c < I^2$, and $R < k$.

Now $M_0^2 \geq 4kI^2T^2$ and $M_0^2 \geq k^2 I^4$. Since $k > 64$,

$$AM_0^2 > \frac{k}{4} A_0^2 c^2 I^2 T^2 > 16Dv^2,$$
$$AM_0^2 > 64A_0^4 k^2 I^4 > 16b^2.$$

Adding these strict inequalities yields $AM_0^2 > 8P_0$. Because $d = Q - 1 \geq Q/2$ and $M = (kI + T)^2 \geq M_0$,

$$\boxed{dAM^2 > 4QP_0.} \tag{12}$$

This estimate uses no root equation. It strengthens the bound used by package 20 and removes its unnecessary factor of two in the inner weight.

4.3 All complex branches have a uniform bound

Factoring $\mathcal{F}$ first with $f^2 = D$ and then with $\tau^2 = A$ gives exactly the coherent branches

$$x = M\tau + y, \qquad \tau^2 = A, \quad f^2 = D, \quad dy^2 = e - B_{\mathrm{aux}} f = Q(vf - b)^2 - 1. \tag{13}$$

This factorization is valid over an algebraic closure, including repeated roots. It does not introduce assumptions on independent square classes. Every branch, including a possible imaginary y, satisfies

$$|y|^2 \leq \frac{2QP_0 + 1}{d} < \frac{4QP_0}{d} < AM^2. \tag{14}$$

It also satisfies the useful separate estimate

$$|y| < 2(v\sqrt{D} + b) < 8k^2 YIT + 10k^3 I^2.$$

Indeed $\sqrt{D} < A_0 I$, $c < 2kY$, $A_0 < 2E < 2k$, and $b < 2kY + 4k^3 I^2 < 5k^3 I^2$ prove the last inequality.

Lemma 4.1 (The main radicand has a large square-root gap)**.** *Under* (10)*,* $\sqrt{G} - \sqrt{A} > kY^2$*.*

Proof. Since $\Delta > (a+1)^2$ and $c > kY$,

$$\sqrt{G} > (a+1)c > (E + Y + 1)kY.$$

On the other hand, $(\ell + k/2)^2 - A = k^2/4 - 1 > 0$, hence $\sqrt{A} < \ell + k/2 = EkY + k/2$. Subtraction gives $\sqrt{G} - \sqrt{A} > k(Y^2 + Y - 1/2) > kY^2$. □

Because $M = k^2 I^2 + 2kIT + T^2$ and $Y \geq 4096$, this gap gives

$$M(\sqrt{G} - \sqrt{A}) > k^3 Y^2 I^2 + 2k^2 Y^2 IT$$
$$> 10k^3 I^2 + 8k^2 YIT > |y|.$$

Thus every root of $\mathcal{F}$ obeys

$$\boxed{|x| < M\sqrt{G} < \frac{L\sqrt{G}}{H}.} \tag{15}$$

This is the key saving: the inner weight M is separated from the outer congruence weight $L = 1 + HM$. One does not need the extra large outer multiplier used in package 20, or the repeated outer weight used in 21.

4.4 The outer norm forces the main root to be integral

Suppose $\mathcal{N} = 0$. If G were nonsquare, expressing either factor in (5) as $E_0 + E_1\sqrt{G}$, with integers E_0, E_1, would show that both coefficients vanish: otherwise G is a rational square and therefore an integer square. Both conjugate factors would then vanish. The two corresponding roots $(S - L\sqrt{G})/H$ and $(S + L\sqrt{G})/H$ differ by $2L\sqrt{G}/H$, contradicting (15).

Therefore an integer u with $u^2 = G$ and a rational $x = (S - Lu)/H$ satisfy $\mathcal{F}(x) = 0$. The root u is positive. If it were negative, $S = Lu + Hx < 0$ by (15), whereas $S = b_0 + Hr > 0$.

4.5 Rational reconstruction and Pell saturation

For real x, inequality (12) and $B_{\mathrm{aux}}\sqrt{D} \le QP_0$ give

$$C(x) > 3QP_0 + 1 > B_{\mathrm{aux}}\sqrt{D} > 0.$$

Moreover $\mathcal{F}(0) = (C(0)^2 - DB_{\mathrm{aux}}^2)^2 > 0$. Define

$$V(x) = C(x)^2 + DB_{\mathrm{aux}}^2 - 4d^2x^2M^2A,$$
$$f = -\frac{V(x)}{2C(x)B_{\mathrm{aux}}}, \qquad \tau = \frac{C(x) + B_{\mathrm{aux}}f}{2dxM}, \qquad y = x - M\tau.$$

All denominators are nonzero. The equation $\mathcal{F}(x) = 0$ gives $f^2 = D$, and the ensuing identity $(C + B_{\mathrm{aux}}f)^2 = 4d^2x^2M^2A$ gives $\tau^2 = A$. Both roots are rational numbers with integer squares, so are integers. The remaining rational y satisfies

$$Q(vf - b)^2 - (Q - 1)y^2 = 1. \tag{16}$$

Lemma 4.2 (Rational auxiliary Pell saturation)**.** *If integers $m \ge 2, V$ and rational y satisfy $m^2V^2 - (m^2 - 1)y^2 = 1$, then y is an integer.*

Proof. Write $y = a/b$ in lowest terms with $b > 0$. The equation implies $b^2 \mid m^2 - 1$. Put $D_0 = (m^2 - 1)/b^2$, a positive nonsquare integer. There are integral Pell solutions

$$m^2 - D_0b^2 = 1, \qquad (m|V|)^2 - D_0a^2 = 1.$$

Neither second coordinate is zero. Let $\epsilon = x_1 + y_1\sqrt{D_0} > 1$ be the fundamental integral Pell unit, and write $\epsilon^n = \chi_n + \psi_n\sqrt{D_0}$. For some $j, l \ge 1$, $m = \chi_j$, $b = \psi_j$, $m|V| = \chi_l$, and $|a| = \psi_l$.

The identity $\chi_j \mid \chi_l$ forces l to be an odd multiple of j. To see this, $\epsilon^{2j} \equiv -1$ in the quadratic ring modulo χ_j gives $\chi_{n+2j} \equiv -\chi_n$ and $\chi_{2j-n} \equiv -\chi_n$. Reduce l modulo $2j$ and reflect into $0 \le n \le j$. For $n < j$, $0 < \chi_n < \chi_j$, so divisibility is impossible; hence the reduced index is j. The Pell multiplication formulas now give $\psi_j \mid \psi_l$. Thus $b \mid a$, and coprimality forces $b = 1$. □

Apply the lemma to $m = \Delta I$ and $V = vf - b$ in (16). It yields $y \in \mathbb{Z}$, and then $x = M\tau + y \in \mathbb{Z}$. The special Pell shape is essential; a general rational solution of $dy^2 = e - Bf$ need not be integral.

4.6 Recover the omitted congruence and positive coordinate

Since $L \equiv 1 \pmod{H}$ and $S = b_0 + Hr = Lu + Hx$, $u \equiv b_0 \pmod{H}$. Therefore

$$\gamma = \frac{u - b_0}{H} \in \mathbb{Z}.$$

It is positive. Write $\psi_p(n)$ for the coefficient of $\sqrt{p^2-1}$ in $(p+\sqrt{p^2-1})^n$. The restored first Pell equation implies $k = 2\psi_{2XY^2+1}(n)$ with $n \geq 1$. Since $k > E \geq 16^4$, $n \neq 1$, so $k \geq 4(2XY^2+1) > X$ and $c > k > X$. Because $u > 0$, $u^2 = 1 + \Delta c^2$, and $\Delta > (a+1)^2$,

$$\gamma H = u - ac - X > c - X > 0.$$

Taking absolute values of τ and y is harmless. If $f < 0$, change the signs of f, T together. The inherited signed-quotient normalized compiler then applies, forces $\varepsilon = 1$, and recovers a blank halting tableau. This use of the signed-quotient result is explicit; no signed-γ theorem is being assumed.

Conversely, take a canonical positive normalized compiler witness, with $u = b_0 + H\gamma$. The choice

$$r = \gamma + M(u+\tau) + y > 0 \tag{17}$$

gives $S = Lu + H(M\tau + y)$ and makes a factor of $\mathcal{N}$ vanish. Together with $N_k = N_t = 1$, it proves completeness. This finishes the new arithmetic part of Theorem 1.1.

5 Exact degrees and simpler-proof alternatives

Give each supplied unknown degree one and each fixed port degree zero. The leading-degree ledger is

Expression	Degree	Expression	Degree	Expression	Degree
c	5	a, H	6	Δ	12
ℓ, I	11	A, G	22	D	34
Q, d	46	b	38	P	92
U, e	122	M	24	L, HM	30
S	11	n, V_0	82	V_1	41
C_0	134	C_1	87	Z	238
K_{c0}	256	K_{c1}	215	W_0	268
W_1	221	N_0	536	N_1	489

The unique top contribution to C_0 is $-H^2 e$. It remains the unique largest contribution through the subsequent squares. In particular N_0^2 has degree 1072, while GN_1^2 has degree only 1000. Writing $q_0 = \beta J$ and $k_0 = \eta + \zeta$ gives the uniform certificate

$$\mathcal{N}_{\text{top}} = 2^{32} i^{48} k_0^{96} w^{80} s^{176} q_0^{656} (q_0 - F)^{16}. \tag{18}$$

This is nonzero for every valid $\beta > 0$. The leading forms of N_k, N_t have degrees $7, 2$, respectively, so $\deg \mathcal{P}_{12} = 1081$. More explicitly, the coefficient of

$$J^{677} hz\, i^{48} \eta^{96} w^{81} s^{177}$$

is $2^{32}\beta^{677} > 0$. Thus exact degree is established for the actual coefficient instance, without relying on small diagnostic replacements.

There is a useful proof-length tradeoff. Replace the carrier by $S = Lb_0 + Hr$, leaving $M = (kI+T)^2$ and $L = 1 + HM$ unchanged. This costs one extra multiplication and retains degree 1081. The inner bound gives $|x| < 2M\sqrt{A}$, while both $\sqrt{G} > \sqrt{A}$ and $b_0 > \sqrt{A}$. For the second inequality,

$$a^2k^2Y^2 - A = k^2\big[EY(2Y^2-1) + Y^4\big] - 1 > 0,$$

and $b_0 > akY$. Consequently $(S + L\sqrt{G})/H > 2M\sqrt{A}$, excluding that conjugate directly. This avoids the outer gap estimates. Its product and system cost 117 and 112 operations. A further variant using $M = kI^2 + T^2$ costs 118 and 113. All three variants are emitted and audited.

6 A smaller complete coefficient tuple

The new adapter retains package 19's 943 Wang tiles, occupied binary edge labels, native support, and anchor ruler [7, 8]. It combines three changes: decode the selector count before other fields; place mask-population padding in guaranteed zero cells; and shorten the spatial and clause shifts using exact support checks.

6.1 The fixed tile language

A tile is (west, north, east, south). Use vertical labels $0, 1$ for headless bits, $\mathrm{head}(q, b) = 2(q+1)+b$ for $0 \le q \le 215$, and $L_0 = 434, Q_0 = 435, R_0 = 436, V_0 = 437, H_0 = 438$ for initialization and boundaries. Horizontal labels are $O = 0, HB = 1, PL = 2, PQ = 3, PR = 4, BAD = 5$ and directed signals. Let S_R, S_L be the sorted sets of states targeted by right- and left-moving working actions. They have sizes 199 and 45. Set $\mathrm{right}(q) = 6+\mathrm{index}_{S_R}(q)$ and $\mathrm{left}(q) = 205+\mathrm{index}_{S_L}(q)$.

The finite language is specified by the following table. Each range is finite, and the builder reconstructs it without running the machine.

Tile family	Count
For each working action $(q, b) \mapsto (v, R, t)$ use $(O, \mathrm{head}(q, b), \mathrm{right}(t), v)$; for a left action use $(\mathrm{left}(t), \mathrm{head}(q, b), O, v)$	430
Halt loops $(O, \mathrm{head}(215, b), O, \mathrm{head}(215, b))$	2
Headless stays (O, b, O, b)	2
Receive-right $(\mathrm{right}(q), b, O, \mathrm{head}(q, b))$; receive-left $(O, b, \mathrm{left}(q), \mathrm{head}(q, b))$	488
Finish (O, c, O, H_0), $c \in \{0, 1, \mathrm{head}(215, 0), \mathrm{head}(215, 1)\}$	4
Initialization $(PL, L_0, PL, 0)$, $(PL, Q_0, PQ, 1)$, $(PQ, R_0, PR, \mathrm{head}(93, 0))$, $(PR, R_0, PR, 0)$	4
Reset (HB, H_0, HB, p), $p \in \{L_0, Q_0, R_0\}$	3
Seam (u, V_0, v, V_0), $u \in \{O, HB, PR\}$, $v \in \{O, HB, PL\}$	9
Forbidden $(BAD, 0, O, 0)$	1
Total	943

The start tile $(PL, Q_0, PQ, 1)$ is selector zero and the forbidden tile is selector one; all remaining tiles follow lexicographically. The actual first machine action is $(0, 0) \mapsto (1, R, 93)$. No east color is BAD, so the forbidden tile cannot occur in a global matching tiling.

The inherited language proof shows that a marked periodic tiling exists exactly when blank K215 halts. Initialization phases force finite vertical boundaries around the marked head, directed signals implement each exact machine step, and a later reset row requires a halted head. Conversely a finite halting run admits independently padded widths and heights and matching helical seams. The adapter changes no tile or machine action.

6.2 Literal coefficient definitions

Put

$$\rho = 2^{125}, \qquad L_{\mathrm{cell}} = 15625, \qquad d_{\mathrm{base}} = 125 \cdot 15625 = 5^9, \qquad B = \rho^{L_{\mathrm{cell}}}.$$

For $0 \le j < 10$, let

$$e_0(j) = 1023 + j, \quad e_1(j) = 1013 + j, \quad e_2(j) = 1003 + j, \quad e_3(j) = 983 + j.$$

The native support is

$$\mathcal{E} = \{0, \ldots, 942\} \cup \{e_t(j) : 0 \le t < 4,\ 0 \le j < 10\} \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.$$

It has 989 elements. Let $w_j = 2^{10+2j}$ for $0 \le j < 44$. If tile i has labels x_{it}, encode the occupied label $x_{it} + 512$ in ten bits, and define

$$\begin{aligned} c_i &= 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) w_{10t+j} + \sum_{j=40}^{43} w_j, \quad 0 \le i < 943, \\ c_{e_t(j)} &= w_{10t+j}, \\ (c_{1095}, c_{1096}, c_{2195}, c_{3296}) &= (w_{40}, w_{41}, w_{42}, w_{43}), \\ c_{1063} = c_{1064} &= 0, \qquad \mu = 1022 + \sum_{j=0}^{43} w_j. \end{aligned}$$

Bit extraction here is only part of the bounded definition of fixed coefficients from the finite tile list. It is not an operation in the Diophantine polynomial.

The external padding mask is

$$Z_{\mathrm{pad}} = (2^{123} - 1) \sum_{e=9000}^{9005} \rho^e + (2^{122} - 1)\rho^{9006}.$$

Define

$$\begin{aligned} D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}), \\ D_R &= \rho^{42}, \\ M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{1063}, \end{aligned}$$

$$M_F^{\mathrm{native}} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1013+j} + \rho^{1065+j}) + \rho^{2195} + \rho^{3296} + Z_{\mathrm{pad}}. \tag{19}$$

The complete tuple used throughout this package is

$$\boxed{\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_F^{\mathrm{native}} + B - 1.} \tag{20}$$

Every exponent and summation range is fixed. There is no unbounded coefficient search and no dependence on whether a refutation is found.

7 Why the smaller encoding preserves the compiler

7.1 Mixed-radix decoding must be done in order

At a clean clause target, let n be the number of selected tiles. Native selector digits are Boolean, so $0 \le n \le 943 < 1024$. All other contributions to the clause sum are divisible by 1024; therefore its low ten bits are exactly n, even if higher fields have not yet been decoded. Disjointness from the low mask 1022 forces $n \in \{0, 1\}$.

Only now consider the upper fields. Each is the sum of at most one selected tile bit and one copy bit, or of n and one anchor bit. Its value lies in $\{0, 1, 2\}$, so radix-four fields are carry-free. The mask tests their low bits, allowing precisely 0 and 2. Thus an empty cell has no copies or anchors, and an occupied cell has exactly the selected tile's copy bits and all four anchors.

Conversely, every valid tile encoding and the empty encoding passes. The old uniform ten-bit spacing is unnecessary after the selector count has been established. The new clause mask has $9 + 44 = 53$ set bits. Assuming the upper fields carry-free before decoding n would be an error; the ordered argument above is the substantive mixed-radix change.

7.2 Exact geometry replaces excessive spacing

Use clause targets $T_1 = 4393, T_2 = 5495$, spatial shift $H_c = 42$, and bare shifts $30, 62, 1100, 2200$. The required target identities, for $0 \le j < 10$, are:

Target	D_C source	D_R source	Temporal source
$1013 + j$	$983 + j$	none	$1013 + j$
$1065 + j$	$1003 + j$	$1023 + j$	none
2195	1095	none	2195
3296	1096	none	3296

For every $e \in \mathcal{E}$, these are exact native-basis coefficient identities, including both reflected clause bands. At each clause target T, $[\rho^{T-e}]D_C = c_e$ and $e + 42 \neq T$. The checker exhausts all 69 230 such identities; linearity extends them to every native assignment, including the allowed dummy digits.

The two finite support intersections are

$$(T_1 - \mathcal{E}) \cap (T_2 - \mathcal{E}) = \varnothing \pmod{15625},$$
$$(2195 - \mathcal{E}) \cap (3296 - \mathcal{E}) = \{0\} \pmod{15625}.$$

The first supplies a clean clause band for every binary rotation; the second supplies unique whole-cell anchor alignment after binary alignment. The upper-dummy spill position 1064 already belongs to $\mathcal{E}$. Its unique possible odd spill cannot satisfy both distinct anchor tests. These are the same synchronization arguments as the inherited exact-support compiler. The support span 3296 is less than half the circumference, so no omitted circular case arises.

7.3 Population padding can test guaranteed zero positions

The two clause masks contribute 106 set bits. Twenty overlap tests, two anchor tests, and the low test contribute 23 more. The remaining 860 bits needed to reach 989 are supplied by Z_{pad}.

For an aligned canonical witness, the largest radix exponent in any D_C convolution is $5495 + 3296 = 8791$; the corresponding bound for D_R is $42 + 3296 = 3338$, and temporal native support is at most 3296. All padding positions $9000, \ldots, 9006$ are therefore identically zero. Completeness loses no canonical witness. In soundness the added tests may simply be discarded; adding zero tests cannot create a witness.

The exact population and low-bit invariants are

$$\mathrm{pc}(M_F^{\mathrm{native}}) = 106 + 23 + 860 = 989,$$
$$\mathrm{pc}(M_C) = d_{\mathrm{base}} - 989,$$
$$M_C \equiv 2 \pmod 4, \qquad M_F^{\mathrm{native}} \equiv 4 \pmod 8,$$
$$0 < M_C, M_F^{\mathrm{native}} < B - 1.$$

They preserve the inherited inverse-population theorem and its exact two-bit surplus. The padding need not occur inside the clause band; the proof uses its population and its vanishing on canonical outputs.

7.4 Carries, margins, and five-adic tuning

For the uniform inherited bound

$$\mathrm{mass} = (989 + 2)\left(2\sum_{e \in \mathcal{E}} c_e + 6\right),$$

mass has 118 bits and 4mass + 8 has 120 bits. The largest padding digit has 123 bits, and $2(2^{123}-1)+4 < 2^{125}$. Therefore $\rho = 2^{125}$ exceeds every required uniform carry threshold. Padding is part of the tested mask, not an additional computation summand. All unshifted products and mask positions remain inside the 15625-cell circumference, and the inherited strict-slack estimates remain available.

Because 125 is a power of five, $\rho \equiv 2 \pmod 5$. The clause gap $1102 \equiv 2 \pmod 4$ makes paired reflected-clause contributions cancel modulo five. The four bare shifts give $D_C \equiv 0 \pmod 5$, while $D_R \equiv 4 \pmod 5$. Consequently

$$\boxed{2D_C - D_R \equiv 1 \pmod 5.}$$

The inherited dummy-control argument needs this quantity to be a unit; its particular nonzero residue is immaterial. Independent power-of-five width and height padding therefore still tunes the actual packed rank to the needed temporal stride. This completes the coefficient-adapter argument and the remaining part of Theorem 1.1.

8 Materialized integers and operation-minimal alternatives

The coefficient payloads are deterministic gzip files containing minimal unsigned little-endian integers, with no header in the decompressed bytes. The binding manifest records SHA-256 hashes of both forms. The sizes are:

Port	Bits	Raw bytes	Gzip bytes
β	1,953,125	244,141	273
K_0	1,958,376	244,797	10,170
M_C	1,953,125	244,141	373
M_F	1,953,126	244,141	371
Total		977,220	11,187

The old 943-tile compact tuple occupies 5 312 269 raw bytes, and its largest coefficient has 13 201 251 bits [8]. The new raw size is 81.604% smaller, while the largest bit length is 85.165% smaller. These are actual coefficient-height improvements, not merely better compression of unchanged integers.

8.1 The unchanged 67-operation polynomial, with smaller coefficients

For users prioritizing operation count, the archive includes the inherited fifteen-positive-variable arithmetic DAG, rebound to (20). Use the common interface with additional positive f, τ, y, γ and without r. Set

$$D_{\text{main}} = X + ac + \gamma H, \qquad V = c(Tf-1) - Rf^2, \qquad K_{\text{aux}} = (i\Delta c^2)^2,$$
$$N^{(0)} = \tau^2 - (XY^2)(XY^2+1)k^2,$$
$$N^{(1)} = D_{\text{main}}^2 - \Delta c^2,$$
$$N^{(a)} = K_{\text{aux}}(V^2 - y^2) + y^2,$$
$$N^{(s)} = \Delta f^2 - K_{\text{aux}}.$$

Then

$$\mathcal{P}_{67} = N^{(0)}N^{(1)}N^{(a)}N_kN_tN^{(s)} - \Delta = 0$$

has the same positive existence predicate, degree 155, and 67 operations [5]. Its arithmetic is not a new result here. Its fixed coefficients are smaller by the new adapter.

The archive also includes package 22's unchanged seven-equation system in sixteen positive unknowns, degree 24 and 59 operations, rebound to the same tuple [9]. These two baselines prevent a twelve-variable improvement from being mistaken for an overall operation-count record.

9 Verification, reproducibility, and limits

The Python programs use only the standard library. The delivered package is designed to run after extraction, without relying on the source repository layout, omitted historical files, or network access. From the package root:

```
python3 reproduce.py
make pdf
```

The first command regenerates the arithmetic circuits and coefficient payloads, runs independent audits, and checks the fixed-instance bindings. The second compiles this source using `pdflatex` twice.

9.1 What the checks establish

The arithmetic builder checks each of three variants against an independently implemented eight-component conjugate-product evaluator on 24 exact signed assignments. It checks product and sum-of-squares finalizers, the parity identity, dependency order, liveness, and operation counts. A separate auditor imports no builder and checks the leading-form identities on 32 exact assignments. The uniform proof in Section 5, not the finite sampling, establishes the degree for the actual large coefficients.

The coefficient verifier reads the actual bytes and independently reconstructs the recipe. Its checks include 69 230 exact native-basis identities, all 943 valid tile clauses, rejection of 41 492 individual copy/anchor mutations, rejection of all 44 analogous empty-cell mutations, all 944 possible selector counts, support intersections, population counts, low congruences, and padding support separation. These finite basis identities establish the stated finite layout properties exactly; the arbitrary-tableau transfer is the mathematical argument in Section 7.

The release verifier additionally evaluates the supplied circuits modulo three primes using residues of the actual four integers, and checks the compact direct formula against the emitted circuit. These computations audit binding and transcription. They are not a search for a full solution or evidence concerning the truth of consistency.

9.2 What is inherited and what remains open

The new proof uses the same native compiler, Pell rank lemmas, half-binomial criterion, and K215 calibration as the reviewed packages. Copies of the critical inherited arguments and an immutable provenance ledger accompany the archive. The new work has received a separate mathematical audit within this assistant collaboration, not an external human endorsement. No Lean or Rocq proof of the new norm or coefficient adapter is claimed.

The improvements do not settle any absolute minimization problem. The nine-natural-variable form remains smaller in variable count; the 67-operation and 59-operation baselines remain smaller in arithmetic work. We have not constructed an eleven-positive-variable equation at moderate degree or a new improvement to the signed 26-integer family.

For contemporary context, Sun's September 2026 preprint states a constructive ten-integer-variable representation for every recursively enumerable set [12]. That general theorem gives

a further route to a small signed-variable count. This package does not instantiate its source-polynomial compiler for K215 and does not present an unspecified polynomial as an explicit ten-variable result. Likewise, the concrete Jones reduction used by package 18 assumes a quartic input: feeding it a degree-1081 polynomial without changing and proving the coding bounds would be invalid.

The most direct formalization project is now comparatively local: prove Lemma 4.1, the coherent-root recovery lemma, the scalar norm identity, and the mixed-radix decoder, then compose them with an explicitly formalized native compiler interface. The materialized tuple and full circuits make this a fixed, reviewable target.

A Package map and exact instance contract

Location	Contents
`report.tex`, `report.pdf`	Complete article and compiled version.
`arithmetic/`	Self-contained emitter, nine circuit variants, algebra receipts.
`coefficients/`	Literal K215 table, ordered tiles, complete coefficient recipe, actual compressed integers, independent byte verifier, binding manifest.
`baselines/`	The unchanged 67-operation polynomial and 59-operation system, with complete new bindings.
`proofs/`	Expanded new proofs and independent mathematical reviews.
`upstream/`	Selected inherited proofs and endpoint source, preserved as evidence.
`verification/`	Independent arithmetic, fixed-instance, and release checks.
`provenance/`	Immutable source paths, revision, and byte hashes.
`licenses/`	Repository license and inherited machine notices.

The canonical instance identifier is

`wang943-mixed125-external-padding-5pow9.`

Circuit aliases `beta` and `Bm1` map to the payload `beta`; `K0` and `Kconstant` map to `K`; `MC` maps to `MC`; and `MF` maps to `MF_source`. All four ports must be replaced together. The shifted mask convention is part of the theorem, not an optional renaming. Historical source copies keep their historical bindings; only the explicitly rebound delivered circuits select this instance.

References

[1] ProveIt contributors, *Con(ZFC) in Diophantine form: audited wave-3 comparison*. Pinned collection README.

[2] ProveIt package 20, *Twelve positive unknowns by amalgamating the main Pell root*. Pinned proof.

[3] ProveIt package 21, *A twelve-positive-variable Diophantine equation for Con(ZFC)*. Pinned proof; scalar norm emitter.

[4] ProveIt package 09, retained by package 20, *Root amalgamation and the signed-quotient interface*. Pinned dependency.

[5] ProveIt, retained by package 21, *Independent proof and source audit: the 67-operation blank-input polynomial*. Pinned proof.

[6] ProveIt, retained by package 21, *Blank-input helical native compiler proof*. Pinned proof.

[7] ProveIt package 19, *Materialized four-coefficient Con(ZFC) instance: 943 signal tiles*. Pinned language and compiler proof.

[8] ProveIt package 19, *Recommended compact instance:* $B = 2^{5^{10}}$. Pinned exact-support layout.

[9] ProveIt package 22, *A 59-operation system in 16 positive unknowns*. Pinned proof.

[10] ProveIt package 18, *Three explicit Diophantine forms of ZFC consistency*. Pinned report source; concrete nine-variable section.

[11] ProveIt contributors, *KEndpoint215.blank_halts_iff_zfc_refutation*. Pinned Lean endpoint source.

[12] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/html/2609.01594v2.